

Shashank Singh

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Summary

Full-stack software engineer and final-year IT undergraduate with 3 internships and 3 deployed products. Ships production features across React/Node.js and Python/FastAPI with hands-on ML depth; strong in DSA, REST APIs, and system design.

Education

Vellore Institute of Technology (VIT)	Aug 2023 – Jun 2027 (Expected)
Bachelor of Technology in Information Technology	CGPA: 8.69/10.0

Technical Skills

- **Languages:** Java, Python, C++, JavaScript (ES6+), TypeScript, SQL
- **Frontend:** React.js, Next.js, Tailwind CSS, Three.js, AngularJS, jQuery
- **Backend:** Node.js, Express, FastAPI, REST APIs, JWT Authentication, MongoDB, MySQL, SQLite
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, System Design Fundamentals
- **DevOps & Tools:** Docker, AWS, CI/CD, Git, GitHub, Agile/Scrum, Postman, Figma, VS Code
- **AI/ML & Data:** PyTorch, Scikit-learn, XGBoost, NLP, Generative AI, Apache Spark

Work Experience

KrishAI <i>Software Development Intern (Remote)</i>	May 2026 – Jul 2026
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- Shipped **15+** full-stack features across 6 Agile sprints, clearing frontend/backend bottlenecks.
- Reduced application response time by **~30%** by refactoring core modules for concurrent-load handling.
- Integrated ML models into production workflows, improving data-pipeline efficiency by **~25%**.

Sooru.AI <i>Software Development Intern</i>	Jun 2025 – Oct 2025
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- Developed interactive 3D visualizations with Three.js, cutting load times **15%** and lifting session duration **12%**.
- Modularized the frontend architecture into **20+** reusable components, reducing duplicate code by **25%**.
- Optimized REST API consumption with backend teams, shrinking initial JavaScript payload by **28%**.

Team WellEx – ONGC Udbhav Initiative <i>ML Contributor (Independent, Part-time)</i>	Jul 2026
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- Prototyped regression pipelines (Python, Scikit-learn) across **10+** geological attributes to estimate oil-gain potential with an independent ONGC team.

Projects

HALO-OT: Behavioral Anomaly Detection Platform (Cyber-Physical OT)	GitHub Live Demo
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- Architected a full-stack platform (FastAPI, JS, SQLite) with REST APIs scoring OT logs in real time (**<200ms**), deployed on Render.
- Engineered a multi-model engine (Isolation Forest, Random Forest, sequence autoencoder) at **0.967 precision** and **0.04% FPR**, with per-alert attribution.
- Produced a synthetic log generator: a **102,296-event** labeled dataset across **11** attack types mapped to MITRE ATT&CK for ICS.
- **Tech:** Python, FastAPI, Scikit-learn, JavaScript, SQLite, Render.

NanoScan: End-to-End PCB Defect Detection	GitHub Live API
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- Delivered an end-to-end full-stack app (Next.js, FastAPI, MongoDB) with REST APIs, auth, and CI/CD, cutting manual inspection **~40%**.
- Trained a hybrid CNN-ensemble (EfficientNetB0 + XGBoost) reaching **99.2% accuracy**; integrated Grad-CAM and SHAP for defect localization.
- **Tech:** Next.js, TypeScript, FastAPI, MongoDB, PyTorch, XGBoost.

SafeTrail: Real-Time Travel Safety Platform	GitHub
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- Launched a full-stack platform (Next.js/React, Python) with REST APIs, auth, and CI/CD, serving live risk data to **100+** users.
- Implemented a DBSCAN clustering pipeline over **10k+** safety records to flag high-risk hotspots for real-time navigation.
- **Tech:** Next.js, React, Tailwind CSS, Python, Scikit-learn.

Leadership & Communication

Chairperson IEEE TEMS VIT	Dec 2025 – Present
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Promoted from AI/ML & Design Mentor after leading top-performing workshop teams.

- Direct a **200+**-member chapter, owning timelines and budgets, and lifted student engagement **20%** via peer mentorship.

AI/ML & Design Mentor IEEE TEMS VIT	Jan 2025 – Dec 2025
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- Led weekly workshops on Figma and ML fundamentals, contributing to **3** top-10 campus hackathon finishes.

Certifications

Generative AI (GenAI) — Google Developers / SmartBridge.